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Testing for mean reversion in Bitcoin returns with Gibbs-sampling-augmented randomization

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ABSTRACT

In the present paper, we attempt to verify whether the Bitcoin log-returns are mean reverted in the presence of heteroskedastic disturbances driven by a mixture distribution. To tackle this problem, we use the autoregression test of mean reversion based on the Gibbs-sampling-augmented randomization methodology. In general, our results indicated that Bitcoin is mean averting for different returns horizons, model specifications and for sub-sample periods, which show the explosive characteristic of the Bitcoin in the period of analysis from 2010 to 2019.

1. Introduction

The analysis of mean reversion in financial assets attracts the interest of many investors, market practitioners, and researchers. It has fundamental implications for investment decisions from portfolio selection to pricing of options. For example, a mean reverting stock market suggests that assets are less risky in the long run. The research in mean reversion for stock prices has shown evidence that stock returns are to some extent predictable (see, for example, Nelson, 1976; Fama, 1981, among others). The presence of mean reverting components in asset prices is directly at odds with the Efficient Market Hypothesis, and it may imply pricing irregularities or market manipulation. In a seminal work Fama and French (1988) analyzed autocorrelations of stock returns for increasing holding periods. A mean reverting component of prices can induce strong negative autocorrelation in long-horizon returns. For periods of length between 3 and 5-years, long-term mean reversion was empirically observed using stock returns between 1926 and 1985. More specifically, 3-year returns showed a negative correlation of 25%, while 5-year returns showed a negative correlation of 40%.

However, the autoregression testing of mean reversion that ignores patterns of heteroscedascity is biased towards rejecting the null hypothesis of mean aversion too often (Kim and Nelson, 1998). Kim and Nelson (1998) question the often used assumption of homoscedastic volatility and argue that the significant divergences in mean reversion results are explained by the volatility's time-variation. They used a Bayesian Markov-switching model capturing shifts and turning points in the volatility process, and propose a

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Gibbs-sampling-augmented randomization which is able to obtain the distribution of the null hypothesis of mean aversion in the presence of shifts in the volatility. Their findings indicate that mean aversion in stock prices are more common than previously found in the literature.

Hence, it is not surprising that mean reversion has been tested extensively for many traditional financial assets as well as commodities and over a variety of time periods, using different methods. However, Bitcoin has so far been unexplored, despite representing an important case study, since it is the most traded in the world among cryptocurrencies. We add to the literature on cryptocurrencies by studying the mean reversion of Bitcoin returns.

Cryptocurrencies have recently received a great deal of attention in the literature. In addition, many investors and market practitioners are interested in the behavior of Bitcoin for trading purposes. Studies have observed that Bitcoin returns exhibit at the same time differences and similarities with other financial time series (Dyhrberg, 2016). Particularly, the presence of volatility clustering justifies the use of heteroscedastic models in the empirical studies (see, for example, Katsiampa, 2017; Baur et al., 2018; Chaim and Laurini, 2018, among others). In this regard, the Markov-switching models are useful for capturing shifts and turning points in the volatility process that is difficult to accommodate using GARCH-types models. Several applications (see, for example, Bariviera, 2017; Ardia et al., 2018) suggest the use of regime-switching models and show that the Bitcoin returns exhibit regime changes. Nonetheless, no previous study has utilized this methodology for the mean reversion analysis.

To the best of our knowledge, testing of mean reversion in the Bitcoin returns is not currently explored in the literature (notable exceptions include: Corbet and Katsiampa, 2018). Thus, this paper aims to extend the Bitcoin mean reversion analysis in two different ways. Firstly, Corbet and Katsiampa (2018) find evidence of asymmetric reverting patterns for positive and negative returns. In contrast, we focus on latent regime switching properties of Bitcoin returns. This approach allows for jumps in the unconditional volatility, and it is interesting when analyzing mean reversion of returns in the long-run. Using a flexible 3-state Markov-switching variance model, our paper is based on the autoregression test of mean reversion following Kim and Nelson (1998). Secondly, we analyze multi-period Bitcoin returns, since this is relevant as investors may change their trading strategies according to the Bitcoin's long-term behavior. In anticipation of the results to follow, we report that the 3-state Markov-switching model is identifiable and well-adjusted to the data. Differently to the existing literature, the methodology is suitable to capture long-run dynamics of volatility changes while the Bayesian approach captured the uncertainty in volatility and parameters. In general, the Bitcoin is strongly mean avert for all returns horizons, using alternative models specifications and taking account for sub-sample periods. This result indicates that the Bitcoin returns are unpredictable, especially considering their explosive characteristic from 2010 to 2019.

In this recent literature, a closely related topic to our work focus on testing the Efficient Market Hypothesis (EMH) for cryptocurrencies (Urquhart, 2016; Nadarajah and Chu, 2017; Vidal-Tomás and Ibañez, 2018). Urquhart (2016) was the first author to analyze this form of efficiency in the Bitcoin market, rejecting the null hypothesis of informational efficiency regardless of the test. Nadarajah and Chu (2017) test the EMH not on the Bitcoin returns but using power transformations of daily returns, whose results showed that the transformed Bitcoin returns are actually market efficient. Vidal-Tomás and Ibañez (2018) examines the semi-strong efficiency of Bitcoin in the Bitstamp and Mt.Gox markets, showing how the digital currency responds to monetary policy and Bitcoin events. Recently, Demir et al. (2018) indicate that Economic Policy Uncertainty (EPU) can be used to predict Bitcoin returns. And also those papers related to the explosive behavior like Cheah and Fry (2015), Corbet et al. (2018), among others. Therefore, our analysis contributes not only to the mean reverting literature but also to those papers in Bitcoin related to the efficient market hypothesis and explosive behavior of Bitcoin given the connection of the topics.

The remainder of this paper is organized as follows: Section 2 presents the methodology; Section 3 describes the data and our empirical results; Section 4 concludes.

2. Methodology

2.1. The autoregression test

Among the different approaches to study the mean reversion hypotheses, Fama and French (1988), simply estimate a regression model using multi-period returns, since the estimated slope coefficient is a measure of the serial autocorrelation of the multi-period returns.¹ Denoting p_t the log of stock price in period t , the multi-period return from t to $t + k$ is given by $R_{t,t+k} = p_{t+k} - p_t$. Therefore, the mean reverting process can be characterized by negative estimated values of the slope coefficient in the following regression model:

$$R_{t,t+k} = \alpha_K + \beta_K R_{t-K,t} + \varepsilon_{t,t+k}. \quad (1)$$

Despite multi-period stock returns seen to be characterized by negative autocorrelation over long intervals, many authors have challenged the reliability of Fama and French's results. Testing whether the sample autocorrelation is significantly below zero demands the sampling distribution and the standard error of β_K under the null hypothesis of no mean reversion. Taking this fact, statistical issues such as the autocorrelations downward bias in finite samples or volatility changes over the sample period have been appointed by empirical studies. For example, Kim et al. (1991) re-examine Fama and French's work estimating the unknown distribution of the autoregressive statistics. Using randomization methods, the authors relaxed the normality assumption for stock returns and suggested that significance levels are much lower than the previously reported.

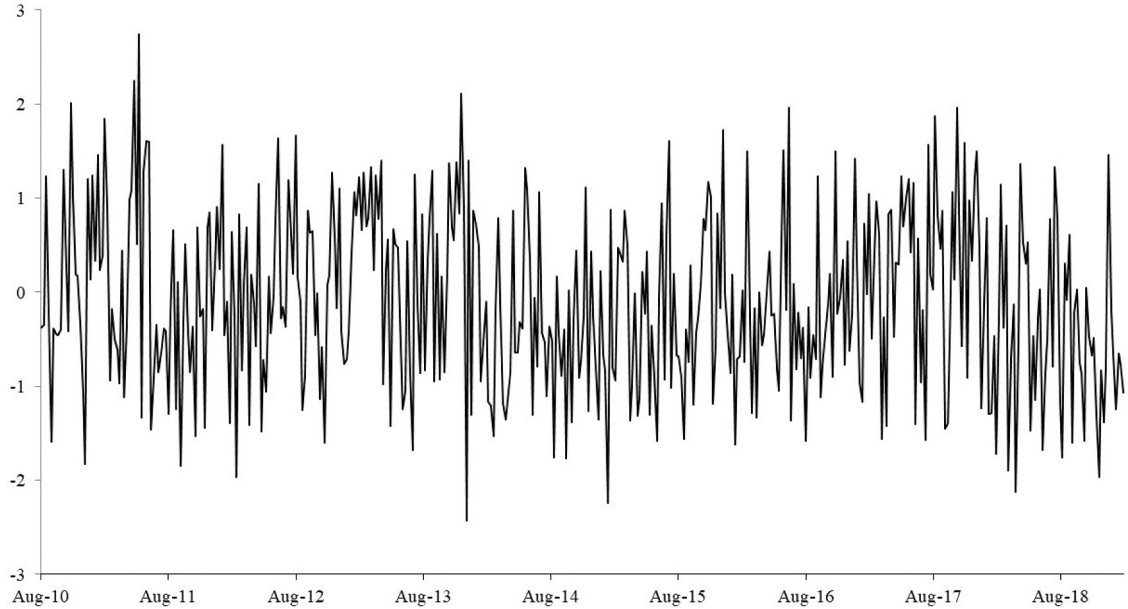
¹ For further details see Cochrane (2001).

Table 1

Summary statistics of the Bitcoin original returns.

Obs.	Mean	Std. Dev.	Min	Max	Skewness	Kurtosis	JB	ARCH (4)
443	0.0247	0.1574	− 0.7223	0.8658	0.6758	8.5413	600.5 (0.000)	116.8 (0.000)

Note: This table presents summary statistics of the weekly original Bitcoin returns from 08/06/2010 to 02/01/2019. Parentheses refer to the p -values of Jarque–Bera test and ARCH test, respectively.

**Fig. 1.** Standardized returns (average of 10,000 realizations from Gibbs-sampling): 3 Regimes Model.**Table 2**

Summary statistics of the simulated standardized returns: 3 Regimes Model.

Obs.	Mean	Std. Dev.	Min	Max	Skewness	Kurtosis	JB	ARCH (4)
443	-0.1044	0.9438	− 2.4336	2.7398	0.1673	2.3789	9.1882 (0.0161)	8.5028 (0.0748)

Note: This table presents summary statistics of the standardized returns (average of 10,000 realizations from Gibbs-sampling). Parentheses refer to the p -values of Jarque–Bera test and ARCH test, respectively.

2.2. Bayesian approach to autoregression test

In the same spirit of [Kim et al. \(1991\)](#), [Kim and Nelson \(1998\)](#) offer an alternative approach to address the issues of heteroscedasticity. Based on previous results in the literature (see, [Kim et al., 1998](#)) the authors apply the Gibbs-sampling-augmented-randomization method in the autoregressive test of mean reversion. In their application, a three-state Markov-switching variance model² is used to approximate the data generating process under the null hypothesis, such as:

$$y_t \sim N(0, \sigma_t^2), \quad (2)$$

$$\sigma_t^2 = \sigma_1^2 S_{1t} + \sigma_2^2 S_{2t} + \sigma_3^2 S_{3t}, \quad (3)$$

$$S_{kt} = \begin{cases} 1 & \text{if } S_t = k \\ 0, & \text{otherwise} \end{cases}, \quad k = 1, 2, 3 \quad (4)$$

$$\sigma_1^2 < \sigma_2^2 < \sigma_3^2, \quad (5)$$

where y_t is the one-period stock log-return subtracted by the mean (i.e., demeaned log-return), and S_t is an latent state variable

² See [Hamilton \(1989\)](#).

Table 3

Bayesian inferences on parameters estimates: 3 Regimes Model.

Parameter	Posterior Mean	Median	SD	95% interval
σ_1^2	0.0018	0.0017	0.0006	(0.0011, 0.0029)
σ_2^2	0.0124	0.0122	0.0023	(0.0093, 0.0163)
σ_3^2	0.0990	0.0955	0.0238	(0.0683, 0.1426)
p_{11}	0.8296	0.8331	0.0503	(0.7418, 0.9059)
p_{12}	0.1345	0.1307	0.0474	(0.0625, 0.2172)
p_{21}	0.0747	0.0709	0.0299	(0.0332, 0.1299)
p_{22}	0.8843	0.8880	0.0332	(0.8237, 0.9321)
p_{31}	0.0500	0.0442	0.0313	(0.0103, 0.1104)
p_{32}	0.0985	0.0926	0.0464	(0.0333, 0.1828)

Note: Non-informative priors were given for all parameters of the model. SD refers to standard deviation.

Table 4

Autoregression tests based on standardized returns: 3 Regimes Model.

Lag K (weeks)												
4	8	12	16	20	24	28	32	36	40	44	48	52
<i>Posterior distribution of $\hat{\beta}_K$ for standardized Bitcoin returns (Mean, Median, SD)</i>												
0.1388	0.1493	0.1413	0.1762	0.2695	0.3100	0.2839	0.1893	0.0986	0.0084	−0.0620	−0.1045	−0.1252
0.1388	0.1501	0.1420	0.1776	0.2708	0.3114	0.2847	0.1902	0.1006	0.0110	−0.0585	−0.1017	−0.1218
0.0316	0.0461	0.0592	0.0661	0.0582	0.0546	0.0580	0.0685	0.0764	0.0816	0.0855	0.0900	0.0926
<i>Sampling distribution. of $\hat{\beta}_K$ based on standardized and randomized returns (Mean, Median, SD)</i>												
−0.0115	−0.0245	−0.0374	−0.0520	−0.0662	−0.0771	−0.0901	−0.1065	−0.1222	−0.1366	−0.1585	−0.1684	−0.1841
−0.0104	−0.0255	−0.0387	−0.0523	−0.0674	−0.0803	−0.0942	−0.1144	−0.1270	−0.1448	−0.1717	−0.1820	−0.1964
0.0782	0.1104	0.1309	0.1552	0.1726	0.1880	0.2021	0.2140	0.2243	0.2387	0.2473	0.2568	0.2674
<i>p-values</i>												
0.9621	0.9242	0.8934	0.9105	0.9653	0.9740	0.9592	0.9050	0.8170	0.7185	0.6525	0.6049	0.5912

Table 5

Autoregression tests based on standardized returns, 2010–2016: 3 Regimes Model.

Lag K (weeks)												
4	8	12	16	20	24	28	32	36	40	44	48	52
<i>Posterior distribution of $\hat{\beta}_K$ for standardized Bitcoin returns (Mean, Median, SD)</i>												
0.1619	0.0787	0.0391	0.0941	0.2331	0.3311	0.3544	0.3356	0.2846	0.2179	0.1498	0.1069	0.0788
0.1625	0.0799	0.0388	0.0945	0.2353	0.3342	0.3578	0.3384	0.2876	0.2203	0.1513	0.1079	0.0802
0.0413	0.0528	0.0688	0.0833	0.0745	0.0683	0.0715	0.0783	0.0840	0.0891	0.0921	0.0973	0.1023
<i>Sampling distribution. of $\hat{\beta}_K$ based on standardized and randomized returns (Mean, Median, SD)</i>												
−0.0156	−0.0328	−0.0485	−0.0692	−0.0886	−0.1091	−0.1295	−0.1500	−0.1684	−0.1852	−0.2075	−0.2381	−0.2571
−0.0156	−0.0334	−0.0494	−0.0707	−0.0893	−0.1155	−0.1354	−0.1577	−0.1808	−0.1979	−0.2198	−0.2525	−0.2779
0.0911	0.1268	0.1535	0.1781	0.1968	0.2150	0.2294	0.2438	0.2575	0.2662	0.2812	0.2907	0.3043
<i>p-values</i>												
0.9635	0.7950	0.6983	0.7973	0.9326	0.9712	0.9754	0.9674	0.9484	0.9181	0.8814	0.8700	0.8493

following an first-order Markov process with transition probabilities given by:

$$Pr[S_t = j | S_{t-1} = i] = p_{ij}, \quad i, j = 1, 2, 3 \quad (6)$$

$$\sum_{j=1}^3 p_{ij} = 1. \quad (7)$$

Using Bayesian methods, the authors estimated the above model applying the Gibbs-sampling technique, which enable us to access the posterior distributions of the parameters σ_1^2 , σ_2^2 , σ_3^2 , p_{ij} ; $i = 1, 2, 3$; $j = 1, 2, 3$, and the latent states S_t , $t = 1, 2, \dots, T$. Theoretically, the estimation procedure provides the uncertainty associated with the parameters and states through the posterior distributions. For example, the r th run of the Gibbs Sampling produce the simulated parameters, $\sigma_1^2(r)$, $\sigma_2^2(r)$, $\sigma_3^2(r)$, and the states $S_t(r)$, $t = 1, 2, \dots, T$ that allow us to generate $\sigma_t^2(r)$, $t = 1, 2, \dots, T$ following Eq. (3). Based on Kim and Nelson (1998), the subsequent steps describe the Gibbs-sampling-augmented randomization method in the autoregressive tests of mean reversion:

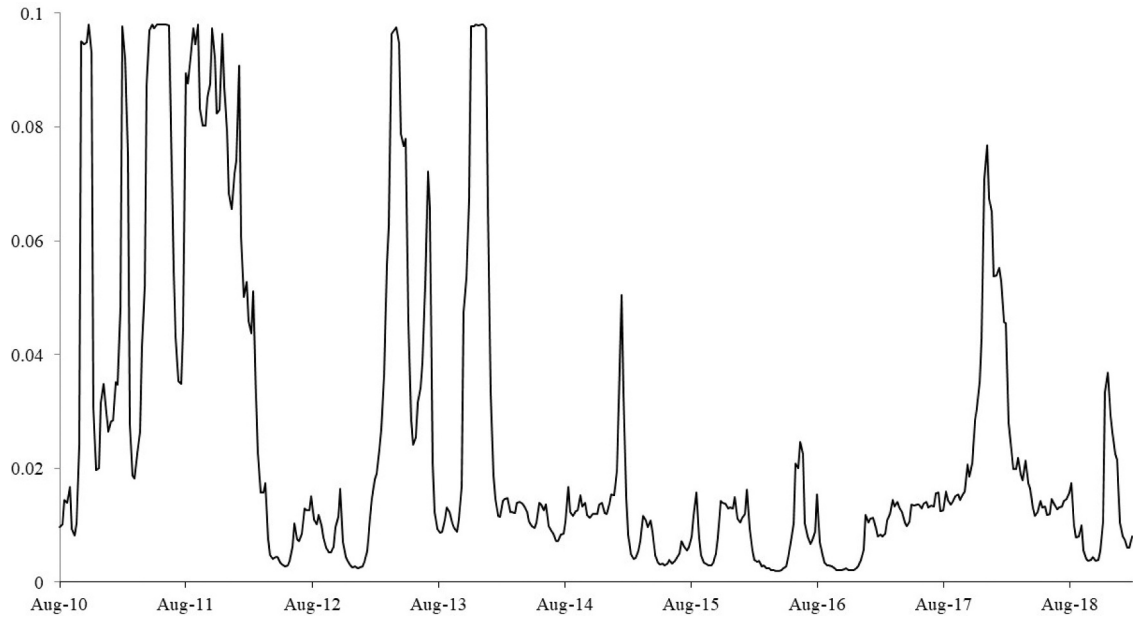


Fig. 2. Estimated variance of Bitcoin returns: 3 Regimes Model.

- Step 1 Standardize the demeaned log-return y_t , $t = 1, 2, \dots, T$, with the simulated $\sigma_t(r)$, $t = 1, 2, \dots, T$ at the r th run of Gibbs-sampling.
- Step 2 Calculate the autoregression coefficients $\hat{\beta}_K^*(r)$ using the standardized returns.
- Step 3 Randomize the standardized returns from Step 1 and calculate the autoregression coefficients $\hat{\beta}_K^{**}(r)$ that are both standardized and randomized.
- Step 4 Compare $\hat{\beta}_K^*(r)$ and $\hat{\beta}_K^{**}(r)$.

The steps 1 to 4 are repeated 10,000 times. Steps 1 and 2 refers to the posterior distribution of the autoregression coefficients for standardized returns. Step 3 refers to the posterior distribution of the autoregression coefficients for standardized under the null hypothesis, i.e., no mean reversion. We count how many times $\hat{\beta}_K^*(r)$ falls below $\hat{\beta}_K^{**}(r)$ to estimate the significance level of the tests.

3. Empirical results

3.1. Data

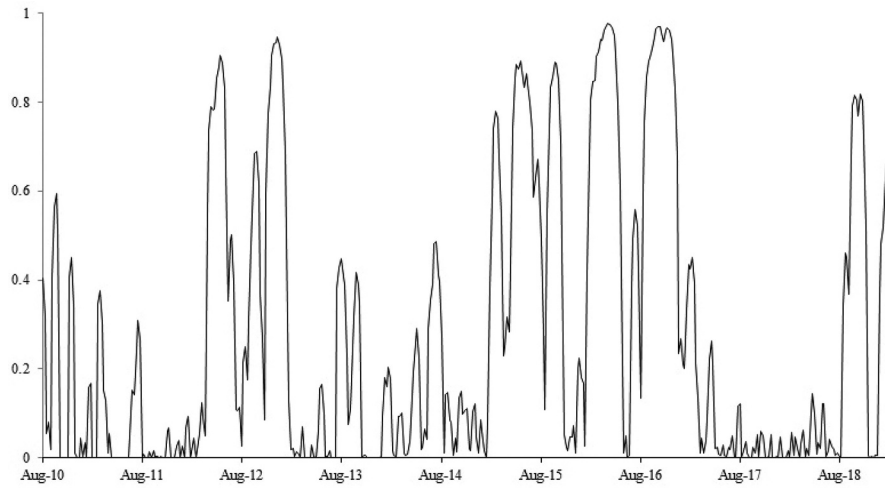
The dataset analyzed in this paper is the weekly closing prices for the Bitcoin (in U.S. dollar) from October 2010 to January 2019 (443 observations) obtained from Bloomberg. The choice of using weekly data takes into consideration a trade-off between the number of observations available for the analysis and the presence of noise regarding the regime switching identification. The use of daily data ultimately confuses the detection of distinguishable regimes. However, monthly data would considerably reduce the number of observations to run the autoregressive test for multi-periods returns. The descriptive statistics for the Bitcoin log-returns are reported in Table 1. The mean is 0.0247 with a standard deviation of 0.1574. The skewness is positive, and the kurtosis is higher than 3. The Jarque–Bera (JB) statistics also indicated the departure from normality, while the value of the ARCH (4) suggests the presence of ARCH effects in the log-returns.

3.2. Estimation results

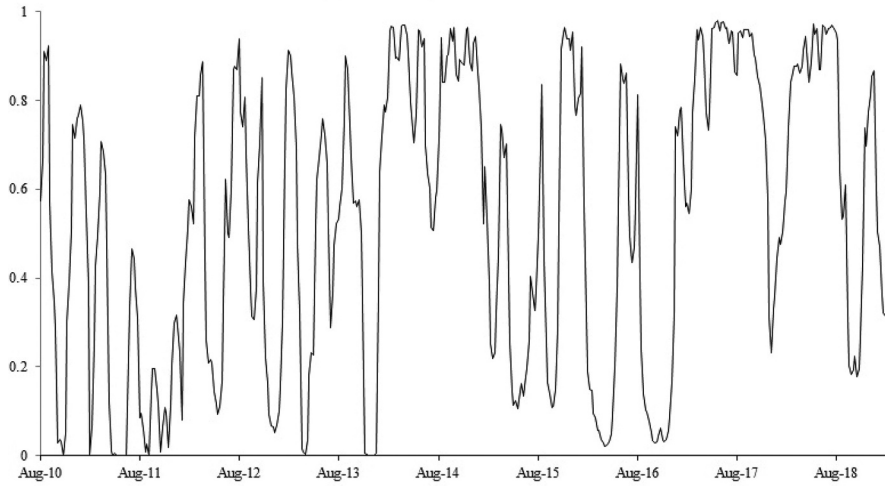
We first evaluate if the three-state Markov-switching model captures most of the dynamics in Bitcoin returns variance following the approach of Kim et al. (1998). For example, Fig. 1 plots the average of 10,000 realizations of the standardized returns described in the Gibbs sampling-augmented randomization method.³ Analyzing the summary statistics of the simulated standardized returns (see Table 2), we observe no ARCH effect at 5% significance level, the kurtosis value is close to the normal distribution and the Jarque–Bera test does not reject the null at 1% significance level.

Table 3 lists the estimates of variance and transition probability parameters for the Markov-switching model. In our analysis, the

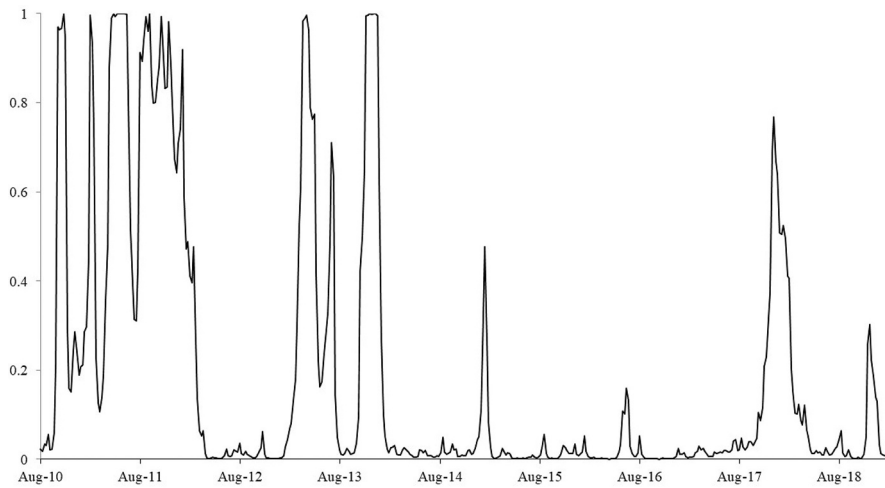
³ We also average the variance series when all Gibbs-sampling interactions over, see Fig. 2 in the Appendix.



(a) Probability low-variance state



(b) Probability medium-variance state



(c) Probability high-variance state

Fig. 3. Probabilities for the three-regime model.

Gibbs-sampling iterates on sampling from the following conditional densities: $S|\Sigma, P$; $\Sigma|P, S$, and $P|\Sigma, S$, where, $\Sigma = \{\sigma_1^2, \sigma_2^2, \sigma_3^2\}$, $P = \{p_{ij}; i, j = 1, 2, 3\}$, and $S = \{S_t, t = 1, 2, \dots, T\}$ are the blocks of parameters. Draws of the variance and transition probability can be sampled from the conditional Inverse-gamma and Dirichlet posteriors distributions, respectively. The state variables are sampled using the so-called forward and backward smoother (see, Chib, 1996). Using non-informative priors for all models parameters, we identify distinct variance values, characterizing low, medium and high volatility states in Bitcoin returns. We also computed the models smoothed probabilities; the regimes are persistent and distinct for which level of volatility, this result is in line with the transition probabilities values, see Fig. 3 in the Appendix.

The autoregressive test based on the Gibbs-sampling-augmented randomization method is reported in Table 4. In our analysis, we considered several multi-period lag returns ranging from 4 weeks (one month) to 52 weeks, approximately one year. In general, our results indicated very strong evidence of mean aversion, since the slope coefficients for the standardized returns are positive. The lowest p -value for the test of mean aversion is 0.026 at lag 24. For long periods, 44, 48 and 52 weeks, the slope coefficients for the standardized returns are negative, but not statistically significant for mean reversion.

We also investigated our preliminary results using two different approaches. First, we re-estimate the Markov-switching model and the autoregressive test using the sub-sample period from 2010 to 2016. Second, we applied the autoregressive test using a two-state Markov-switching model and tested if the model specification would considerably affect its results. Recently, empirical studies have suggested that Bitcoin entered in a consistent bubble-phase since 2017. Before 2017, Bitcoin prices remained bounded around U \$1,000, but after this period, prices sharply increased breaking the value of U\$17,000 in December of 2017, and so far has remained well above the values registered in 2010–2016.

For the sub-sample period 2010–2016, (see, Table 5) we also obtained similar results indicating mean aversion for the Bitcoin returns. In this case, the slope coefficients for the standardized returns are positive for all lags returns. The lowest p -value for the test of mean aversion is 0.024 at lag 28 and 0.205 for the highest at lag 8. However, most of lags presented p -value inferior of 10%. For the two-state Markov-switching model, we also obtained similar results to the previous autoregressive test, both for full sample and sub-sample periods (see Tables 6 and 7 in the Appendix section).

Despite the extensive research devoted to the Bitcoin market, the number of papers examining the mean reversion hypotheses is relatively small. For the best of our knowledge, Corbet and Katsiampa (2018) found evidence of asymmetric reverting patterns in the Bitcoin price returns using ANAR models. Differently to our approach, the authors concluded that negative price returns present stronger reverting behaviors compared to positive returns. However, this reverting pattern becomes more symmetrical for lower data frequencies. Further extension of the present paper could incorporate alternative models of mean reversion, since our results contrast to the existing in the empirical literature. Taking this fact, one readily extension of this work is the implementation of the variance-ratio test using the Markov-switching model as seen in Kim et al. (1998). Another further interesting extension is to analyze mean reversion of different cryptocurrencies, given the increasing literature of the efficient market hypothesis using different cryptocurrencies (Wei, 2018; Vidal-Tomá s et al., 2019; Bouri et al., 2019).

4. Conclusion

In this paper, we captured different states in Bitcoin volatility returns using a 3-state Bayesian Markov-switching model, and use the Gibbs-sampling-augmented randomization method in the autoregressive test of mean reversion for multi-period returns. For the sample period from 2010 to 2019, we found strong evidence of mean aversion using weekly data. We also confirmed our preliminary results for different models specifications and also carrying the analysis for sub-sample periods.

In general, our results indicate that Bitcoin returns do not display evidence of mean reverting components either in the medium or long-run. This implies that the they are unpredictable, and their movement could be explained by random innovations, supporting the hypothesis of market efficiency. In this context winning Bitcoin portfolios are expected to outperform losing portfolios in the long-run.

Appendix

Table 6
Autoregression tests based on standardized returns: 2 Regimes Model.

Lag K (weeks)												
4	8	12	16	20	24	28	32	36	40	44	48	52
<i>Posterior distribution of $\hat{\beta}_K$ for standardized Bitcoin returns (Mean, Median, SD)</i>												
0.1761	0.1421	0.1043	0.1392	0.2558	0.3072	0.2873	0.1967	0.1096	0.0213	−0.0481	−0.0879	−0.1066
0.1763	0.1436	0.1055	0.1418	0.2591	0.3090	0.2890	0.1985	0.1112	0.0225	−0.0470	−0.0868	−0.1055
0.0334	0.0373	0.0443	0.0491	0.0418	0.0369	0.0393	0.0419	0.0431	0.0439	0.0450	0.0461	0.0469
<i>Sampling distribution of $\hat{\beta}_K$ based on standardized and randomized returns (Mean, Median, SD)</i>												
−0.0122	−0.0249	−0.0348	−0.0486	−0.0624	−0.0759	−0.0899	−0.1086	−0.1232	−0.1352	−0.1573	−0.1677	−0.1888
−0.0114	−0.0242	−0.0353	−0.0492	−0.0641	−0.0786	−0.0931	−0.1113	−0.1292	−0.1414	−0.1639	−0.1807	−0.1994
0.0780	0.1102	0.1348	0.1542	0.1725	0.1871	0.2012	0.2157	0.2258	0.2392	0.2491	0.2596	0.2651
<i>p-values</i>												
0.9866	0.9233	0.8377	0.8760	0.9606	0.9772	0.9645	0.9115	0.8360	0.7377	0.6698	0.6273	0.6314

Table 7

Autoregression tests based on standardized returns, 2010–2016: 2 Regimes Model.

Lag K (weeks)												
4	8	12	16	20	24	28	32	36	40	44	48	52
<i>Posterior distribution of $\hat{\beta}_K$ for standardized Bitcoin returns (Mean, Median, SD)</i>												
0.2178	0.1044	0.0146	0.0663	0.2308	0.3409	0.3588	0.3199	0.2583	0.1864	0.1137	0.0646	0.0323
0.2194	0.1067	0.0162	0.0705	0.2361	0.3448	0.3623	0.3227	0.2605	0.1876	0.1140	0.0643	0.0312
0.0375	0.0444	0.0523	0.0618	0.0547	0.0481	0.0493	0.0472	0.0451	0.0443	0.0444	0.0488	0.0554
<i>Sampling distribution of $\hat{\beta}_K$ based on standardized and randomized returns (Mean, Median, SD)</i>												
-0.0147	-0.0320	-0.0484	-0.0676	-0.0874	-0.1076	-0.1264	-0.1469	-0.1695	-0.1882	-0.2145	-0.2367	-0.2569
-0.0149	-0.0316	-0.0512	-0.0685	-0.0901	-0.1120	-0.1304	-0.1548	-0.1791	-0.2050	-0.2278	-0.2551	-0.2770
0.0904	0.1276	0.1551	0.1776	0.1955	0.2115	0.2312	0.2435	0.2555	0.2699	0.2779	0.2923	0.3032
<i>p-values</i>												
0.9896	0.8426	0.6471	0.7603	0.9389	0.9777	0.9786	0.9647	0.9443	0.9052	0.8689	0.8411	0.8307

Supplementary material

Supplementary material associated with this article can be found, in the online version, at [10.1016/j.frl.2019.07.025](https://doi.org/10.1016/j.frl.2019.07.025).

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